

Name _____ Date _____ Period _____

Parent Functions III: $1/x$ and Asymptotes - Practice

Algebra II | Unit 1 | Lesson 6 Practice | TEK A2.2.A

Objective: I can evaluate and solve with the reciprocal function, and describe its asymptotic behavior.

Section 1 — Recall

1. Evaluate: $1/5 =$ _____ $1/(-4) =$ _____ $1/(1/8) =$ _____

2. Is $x = 0$ in the domain of $f(x) = 1/x$? Why or why not?

Section 2 — Apply

3. Solve $1/x = 8$, and $1/x = -1/3$.

$x =$ _____ $x =$ _____

4. As x grows very large and positive, what happens to $1/x$? As x approaches 0 from the right, what happens to $1/x$?

Large positive x : _____ Near 0+: _____

Section 3 — Challenge

5. A function has a vertical asymptote at $x = 0$, a horizontal asymptote at $y = 0$, two branches, and passes through $(1, 1)$ and $(-1, -1)$. Name it, and explain which quadrants it occupies.

Name: _____ Quadrants: _____

6. True or false, with justification: "Since $1/x$ never touches $y = 0$, no function can ever cross a horizontal asymptote."

Section 4 — Explain It

7. A classmate says every parent function with a restricted domain must also have an asymptote, "since that's why the domain is restricted." Is the classmate correct? Test using $\sqrt{x}$.
